

M_E = Mass of Earth = 5.9722×10^{24} Kg

M_S = Mass of Sun = 1.989×10^{30} Kg

$u = GM_S = 1.3271244 \times 10^{20}\ m^3/s^2$.

r_1 = Initial orbital radius = 1.496×10^{11} m.

$r_2 = 2.7 r_1 = 4.0392 \times 10^{11}$ m

c = Speed of Light = 3×10^8 m/s.

T = 10000 years = $10000 \times 365 \times 24 \times 3600$
$= 3.15 \times 10^{11}$ s.

Earth's initial orbital spin, V_o

$$F_g = \frac{GM_E M_S}{r^2} \qquad F_c = \frac{M_E v^2}{r}$$

Now $F_g = F_c$

$$v^2 = \frac{GM_S}{r} \Rightarrow \boxed{v_{o1}^2 = \frac{u}{r_1}}$$

$$v_{o1} = \sqrt{\frac{1.3271244 \times 10^{20}}{1.496 \times 10^{11}}}\ m/s.$$

$$= \sqrt{0.8871152 \times 10^9}$$

$$= \sqrt{8.871 \times 10^8}\ m/s$$

$$= 2.978 \times 10^4 = 29780\ m/s = 29.78\ Km/s.$$

$$v_{o2} = \sqrt{\frac{1.3271244 \times 10^{20}}{2.7 \times 1.496 \times 10^{11}}}$$

$$= \frac{2.978 \times 10^4}{\sqrt{2.7}} = \frac{2.978 \times 10^4}{1.6431} = 18120\ m/s$$

$$= 18.12\ Km/s$$

Hohmann transfer

semimajor axis $\Rightarrow A = \frac{r_1 + r_2}{2} = 1.85 r_1$

$= 1.85 \times 1.496 \times 10^{11}$

$= 2.76676 \times 10^{11}$ m

Ellips start velocity at r_1

$$v_{e1} = \sqrt{u\left(\frac{2}{r_1} - \frac{1}{a}\right)}$$

$$= \sqrt{1.32712744 \times 10^{20}\left(\frac{2}{1.496 \times 10^{11}} - \frac{1}{2.76676 \times 10^{11}}\right)}$$

$= 35980$ m/s $= 35.98$ km/s.

$$v_{e2} = \sqrt{u\left(\frac{2}{r_2} - \frac{1}{a}\right)}$$

$$= \sqrt{1.32712744 \times 10^{20}\left(\frac{2}{2.7 \times 1.496 \times 10^{11}} - \frac{1}{2.76676 \times 10^{11}}\right)}$$

$= 13340$ m/s $= 13.34$ km/s.

$\Delta V_1 = 35.98$ Km/s $- 29.785$ km/s

$= 6.195$ km/s.

$\Delta V_2 = 18.13$ km/s $- 13.34$

$= 4.79$ km/s.

Energy $\varepsilon = -\frac{u M_E}{2a}$

$\varepsilon_1 = -\frac{u M_E}{2 r_1}$ $\qquad$ $\varepsilon_2 = -\frac{u M_E}{2 r_2}$

Required energy $= \frac{u M_E}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

$= 1.32712744 \times 5.9722 \times 10^{24} \times 10^{20} \times \left\{\frac{1}{1.496 \times 10^{11}} - \frac{1}{4.039 \times 10^{11}}\right\}$

required $= 1.67 \times 10^{33}$ J

Momentum transfer $\Rightarrow M_E \Delta v$

$$\Delta P_1 = M_e \Delta V_1$$
$$= 5.9722 \times 10^{24} \times 6.29 \times 10^3$$
$$= 36.967 \times 10^{27} \text{ Kg m/s}$$

$$\Delta P_2 = M_E \Delta V_2$$
$$= 5.9722 \times 10^{24} \times 4.79 \times 10^3$$
$$= 28.6068 \times 10^{27} \text{ Kg m/s}$$

Now $F_{req} = \dfrac{\Delta P_1 + \Delta P_2}{T}$

$$= \frac{65.5738 \times 10^{27}}{3.15 \times 10^{11}}$$

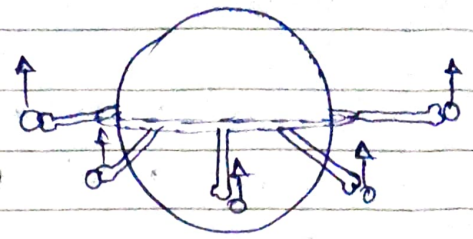
$$= \boxed{2.08 \times 10^{17} \text{ N}}$$

$p_n = \dfrac{\text{Energy of photon}}{c}$

→ **Radius of Arm To Bear Throwing Velocity of Asteroid :**

$$\delta F = \frac{Mv^2}{\text{dist.}}$$

$$\delta F = \frac{Mv^2}{3R_E}$$



To reach 2.7AU in 10,000 years, ~~radius~~ $v_{req.}$ for shooting of asteroids of 10^{18} Kg bunches at a time, by simulations, $v_{req.} = 50,000$ Km/s.

Now, $$\delta F = \frac{Mv^2}{3R_E}$$

$$\delta \sigma A = \frac{Mv^2}{3R_E}$$

$$\delta \pi \sigma R_{arm}^2 = \frac{Mv^2}{3R_E}$$

Substitute values:

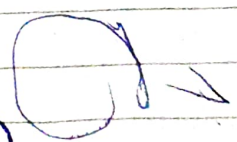
$$R_{arm}^2 (8 \times 3.14 \times 130 \times 10^9) = \frac{10^{18} \times 25 \times 10^{10}}{3 \times 64 \times 10^6}$$

$$R_{arm}^2 = \frac{25 \times 3.14}{8 \times 3 \times 64} \times 10^{15}$$

$$R_{arm} = \frac{50}{16\sqrt{2}} \times 10^{7} \sqrt{5}$$

10^4 000

$R_{arm} \approx 40$Km



R_{arm} taken = 50Km

→ Energy Mining:

① Hydrogen Collecting Disk

1 Km^3 → 10^{15} H_2 Atoms

1 m^3 → 10^5 H Atoms

$8\pi R^2$ Area

$\Delta E = 1.67 \times 10^{33}$ J = $\frac{1}{2} m_e \Delta v^2$

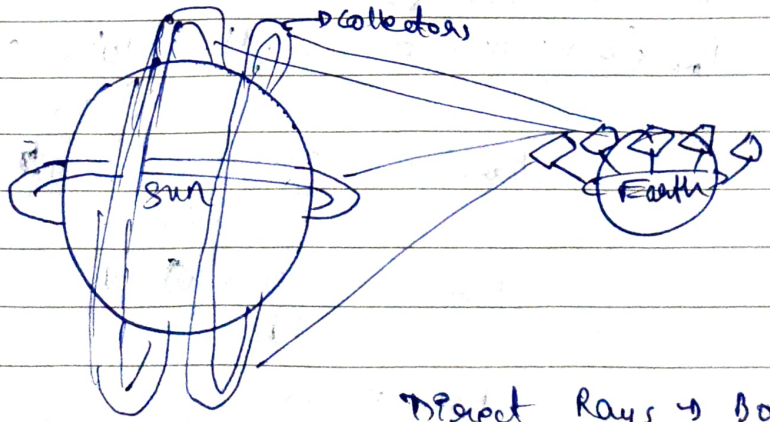
$\Delta v = 2.3 \times 10^4$ m/s → ~~23~~ 23 Kms^{-1}

Vol = (23 Kms^{-1})(10,000 yrs) × $8\pi R^2$ → Infeasible amount of H attached to perform fusion.



② Solar Collecting Disk → ~~Not~~ Input energy produced same che z with solar

③ Make Dyson Sphere / Solar ~~Collectors~~ Enablers



Direct Rays → Boom

We don't need dat much flux, make mirrors with 99 % reflection 1 % absorption

(1385 const.)

Calculated Area of Collector → 59.6 Mil. Km^2

i.e. very Big

→ Momentus Balancing / Gaining :

* Milankowich → $e' < 0.06e$ for life to be sustained

① Propulsor using Earth's fuel can't be done bcz no fuel.

② Catching extraterrestial objects, can be done but need extraterrestial objects.

③ Escavate Earth's mass then throw it out. Acc^n to cal^n need like 4 % of Earth's mass to ~~se~~ throw objects at whopping 1 % speed of light.

④ Use Lorentz force → Obv. doesn't ~~does~~ work in uniform magnetic field. (60N' only)

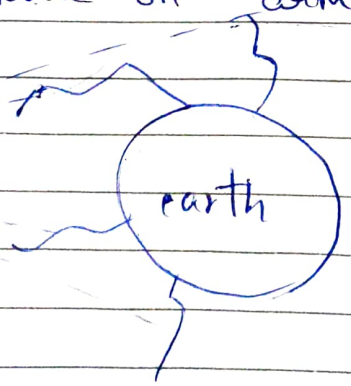
⑤ Laser Mirror

Use photons → get momentum

Force on arm = $\frac{2p}{c}$ = $\frac{2 \times 65.5738 \times 10^{27}}{3 \times 10^8}$

= 45.9×10^{21} N

= 4.59×10^{22} N → F_{mass} on each arm $\approx 4 \times 10^{21}$



So not possible